

AI-Powered Admission Success Predictor

End-to-End Machine Learning Application

PYTHON

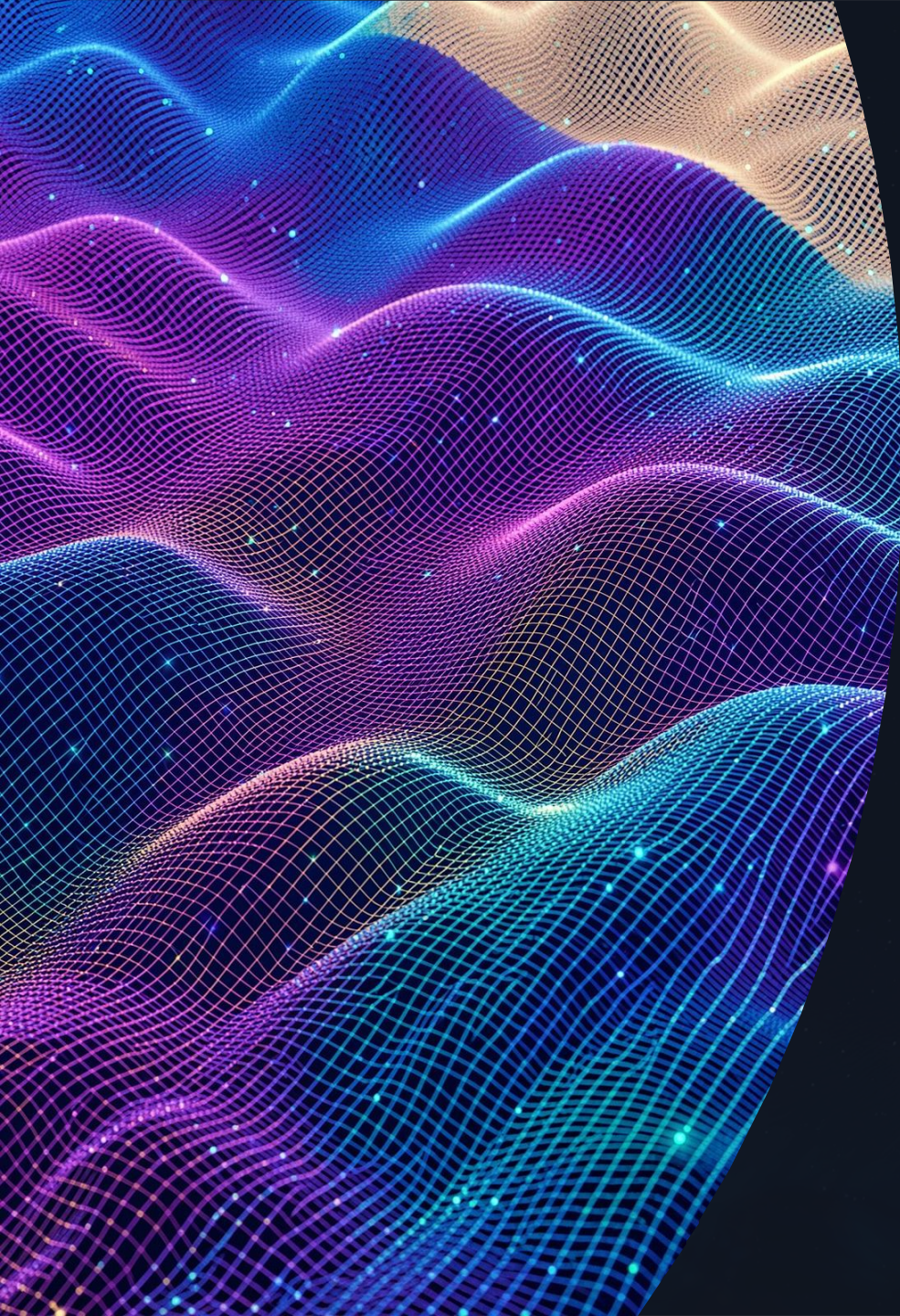
SCIKIT-LEARN

STREAMLIT

GROQ LLAMA 3.3

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Made with **GAMMA**



The Problem

The Challenge

- Admission decisions depend on multiple academic indicators
- Applicants have no visibility into factors influencing decisions
- Difficult to identify how to strengthen applications

Our Solution

- Predict admission probability with high accuracy
- Explain predictions using ML interpretability
- AI-generated personalized guidance via Groq Llama 3.3
- Deploy interactive web application for accessibility

Dataset Overview

Graduate Admission Dataset (Kaggle) — 400 applicant profiles



GRE Score

Graduate Record Exam



University Rating

Institution prestige



LOR

Letter of Recommendation quality



Research

Research experience indicator



TOEFL Score

English proficiency



SOP

Statement of Purpose strength



CGPA

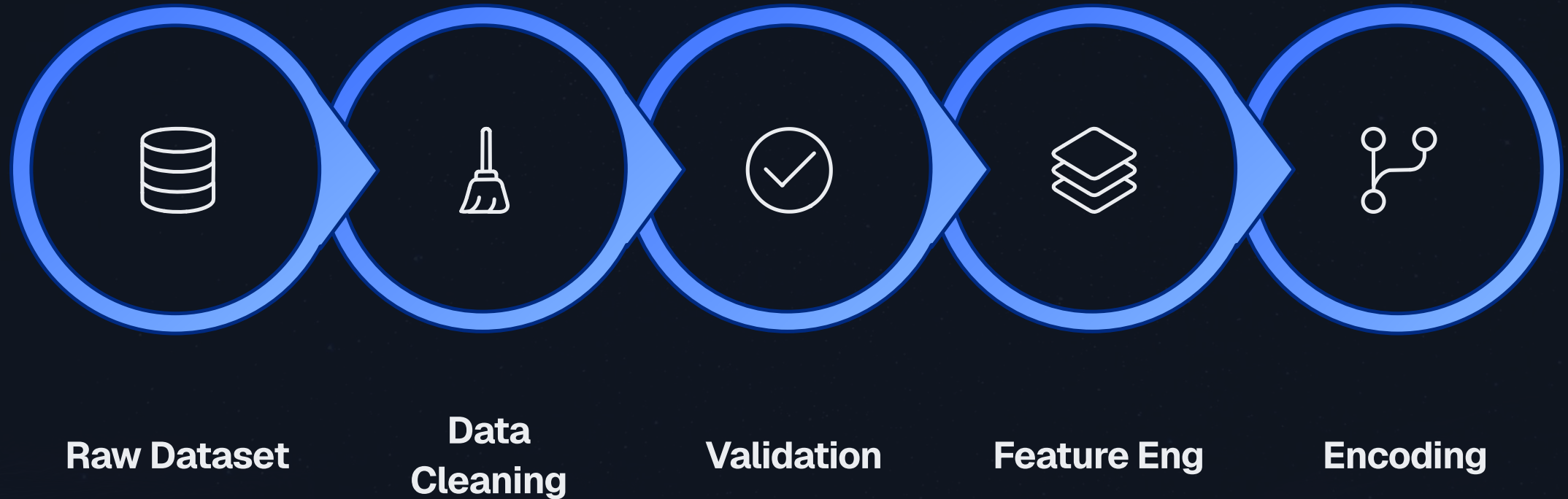
Cumulative GPA



Engineered

Program category • Gender • Completion status

Data Engineering Pipeline



Pipeline automation via **ColumnTransformer** — OneHotEncoder for categorical variables, StandardScaler for numerical normalization, duplicate removal, and missing value handling.

Exploratory Data Analysis

Key Insights

CGPA Impact

Strongest single predictor of admission probability

Research Experience

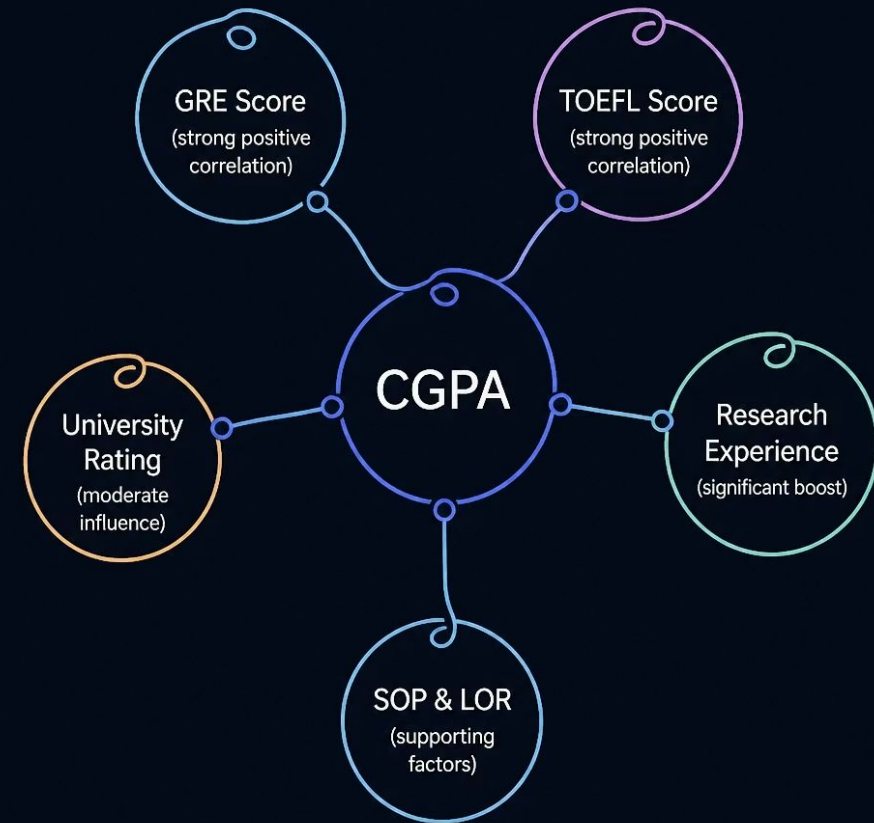
Significantly improves admission chances

GRE & TOEFL

Increase prediction confidence and model robustness

Score Distribution

Most applicants cluster in mid-range scores



Machine Learning Workflow



ColumnTransformer

Heterogeneous data handling

OneHotEncoder

Categorical feature encoding

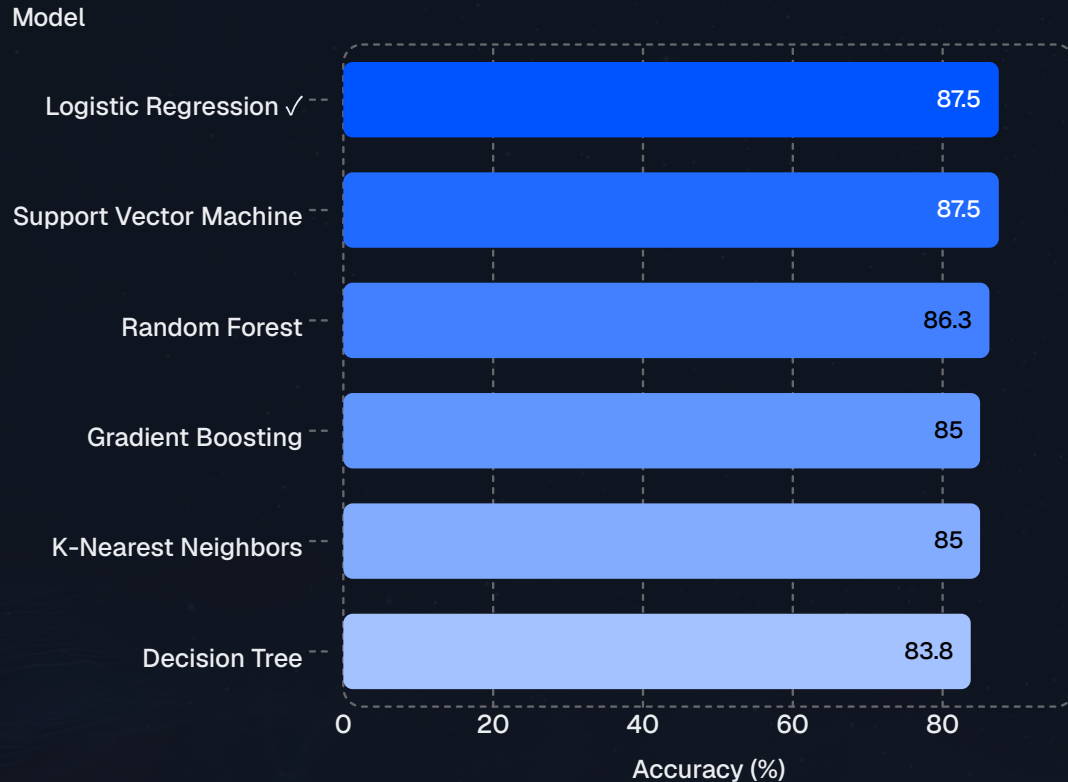
StandardScaler

Numerical normalization

Cross-Validation

Robust performance estimation

Model Comparison & Selection



Why Logistic Regression?

Tied for highest accuracy with SVM

Fast inference for real-time predictions

Highly interpretable coefficients

Strong generalization to unseen data

Best Model Performance

87.50%

Accuracy

89.74%

Precision

85.37%

Recall

87.50%

F1 Score

96.06%

ROC-AUC

High precision minimizes false positives • Strong recall captures true admissions • Excellent ROC-AUC indicates superior discrimination

Visualization Summary



Correlation Heatmap

Relationships between numerical applicant features.

Shows feature interactions



CGPA vs. Admission

Higher CGPA is generally associated with stronger admission likelihood.

Positive relationship observed



Research Experience vs. Admission

Compares admission outcomes for applicants with and without research experience.

Research experience matters



Model Comparison

Compares classifier performance across candidate models.

Logistic Regression and SVM: 87.50% accuracy



Confusion Matrix

Shows correct and incorrect predictions from the selected Logistic Regression model.

Evaluates prediction quality



ROC Curve

Demonstrates the model's ability to distinguish admission outcomes.

ROC-AUC: 96.06%

Challenges

Technical Challenges

Mixed feature type
preprocessing complexity

Feature engineering for
domain knowledge capture

Package compatibility
across Python
environments

Model serialization for
production deployment

API integration with Groq
LLM service

Balancing explainability
with prediction accuracy

Key Takeaways

Preprocessing quality directly impacts model performance

Rigorous evaluation prevents overfitting

Production-ready code requires careful architecture

Combining traditional ML with LLMs creates powerful
systems

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[Live Demo →](#)